Title

Impact of Agricultural Technology Adoption and Socio-Economic Factors on Marginal Farmers in Bihar

Objective

To examine the impact of agricultural technology adoption and socio-economic factors on the productivity and income of marginal farmers in Bihar.

a. To identify the extent and patterns of agricultural technology adoption among marginal farmers.

b. To analyze the socio-economic characteristics influencing technology adoption.

c. To assess the impact of technology adoption on farm productivity

d. To evaluate the effect of technology adoption on household income

Hypotheses

1. Education level of farmers has a positive and significant effect on agricultural technology adoption among marginal farmers in Bihar.
2. Access to credit positively influences the adoption of agricultural technologies.
3. Farm size positively affects the likelihood of technology adoption.
4. Adoption of agricultural technologies has a positive and significant impact on crop productivity.
5. Adoption of agricultural technologies leads to higher household income.

Findings & Analysis

I. Visualization (Charts/Graphs) –

a. The visual analysis suggests a clear positive relationship between technology adoption and farm outcomes.

b. Farmers with higher levels of adoption show noticeably higher crop yields and household income compared to non-adopters

c. The scatter plots further indicate that education is positively associated with income

d. The correlation patterns also support these trends, showing positive associations between technology adoption, income, and key socio-economic factors.

Analysis - Overall, the charts point toward a consistent pattern that better access to resources and knowledge is linked with improved farm performance.

II. Modeling (Linear Regressions) –

a. Education, access to credit, and landholding size significantly increase the likelihood of adopting agricultural technologies among marginal farmers in Bihar.

b. The distance to markets reduces the chances of adoption of agricultural technologies.

c. The technology adoption has a strong and statistically significant positive effect on crop yield, confirming its role in improving productivity.

d. The impact of technology adoption on household income is positive but not statistically significant.

e. Access to credit shows a negative effect on crop yield.

Analysis - The regression results provide a more rigorous assessment of these relationships among marginal farmers in Bihar. While education, access to credit, and landholding size significantly influence the adoption of agricultural technologies, the results confirm that technology adoption has a strong positive impact on crop productivity.

Inference

Taken together, the charts and regression results present a nuanced picture of agricultural outcomes among marginal farmers. Education, access to credit, and landholding size emerge as important determinants of technology adoption. In turn, technology adoption has a significant positive effect on crop yield, highlighting its role in improving productivity.

However, its effect on household income remains statistically insignificant, suggesting that gains in productivity do not necessarily translate into higher earnings for farmers. This indicates a gap between production improvements and income realization, likely influenced by market and structural constraints.

Additionally, the negative effect of credit on yield suggests possible inefficiencies in the utilization of borrowed funds, or that credit constrained farmers may face underlying productivity challenges.

Hypothesis Testing

1. Education level of farmers has a positive and significant effect on agricultural technology adoption among marginal farmers in Bihar.

   Result : Accepted

   Education is found to have a positive and statistically significant effect on agricultural technology adoption, indicating that more educated farmers are more likely to adopt modern technologies.

2. Access to credit positively influences the adoption of agricultural technologies.

   Result : Accepted

   Access to credit shows a positive and significant influence on technology adoption, suggesting that financial support plays an important role in overcoming liquidity constraints.

3. Farm size positively affects the likelihood of technology adoption.

   Result: Accepted

   Farm size has a positive and statistically significant effect on adoption decisions, indicating that relatively larger marginal farms are more willing and able to adopt new technologies.

4. Adoption of agricultural technologies has a positive and significant impact on crop productivity.

   Result: Accepted
   Agricultural technology adoption is found to have a strong and statistically significant positive effect on crop yield, confirming its effectiveness in improving farm productivity.

5. Adoption of agricultural technologies leads to higher household income.

   Result: Rejected statistically
   Although the relationship is positive, it is not statistically significant. This suggests that improvements in productivity due to technology adoption do not necessarily translate into higher household income, possibly due to market constraints, input costs, or price variability.

Policy Recommendations

The findings from marginal farmers in Bihar clearly show that improving productivity is not just about introducing new technology, but also about making it practical, affordable, and easier to use in real farming conditions. Based on this, a few focused policy steps are suggested.

I. **Making modern technology affordable through simple IoT tools -** For most marginal farmers, advanced machinery is often too costly or difficult to use. A more realistic approach is to promote low-cost, easy-to-use IoT-based tools like basic soil sensors, simple weather alerts on mobile phones, and irrigation guidance systems. These don't need to be highly technical, but they can still help farmers make better day-to-day decisions. The key is to keep these tools affordable and locally accessible, possibly through subsidies or community based distribution models.

II. **Building practical, hands-on technology awareness -** Many farmers are still not fully comfortable with modern farming technologies, not because they are unwilling, but because they haven't been properly introduced to them. Regular field-based training through self-help groups, village demonstrations like nukkad natak, and simple digital guidance in local languages can make a real difference. Instead of theory-heavy programs, the focus should be on showing farmers *how* these technologies work in their own fields.

III. **Easier access to safe and affordable institutional credit -** The study also suggests that while credit is important, many farmers may end up depending on informal or high-interest loans, which can actually reduce their ability to invest productively. To address this, farmers should be encouraged to use institutional credit systems like the Kisan Credit Card (KCC). This will only work if the process is made simpler, awareness is increased, and support is provided at the village level so that farmers can access formal credit without difficulty.

In simple terms, farmers do not just need technology, they need technology that fits their reality. When low-cost innovation, practical learning, and safe credit come together, the gap between higher productivity and higher income can be reduced.

Limitations

- Small sample size: The study is based on 140 observations, which limits the scope for broader generalisation across different regions and farming conditions in Bihar.
- Credit measurement issue: The data does not clearly distinguish between subsidized institutional credit (e.g., KCC) and informal or high-interest loans, which may affect the interpretation of credit-related results.
- Technology variable: Technology use (tech_score) is measured on a ranked scale (1, 2, 3), but the categories are not clearly defined in terms of specific technologies or intensity of adoption.
- Cross-sectional data: The study uses one-time data, which does not capture long-term changes or dynamic effects of technology adoption and credit usage.
- Limited depth of variables: Some important factors such as market price variation, input costs, and quality of extension services could not be fully captured in the analysis.

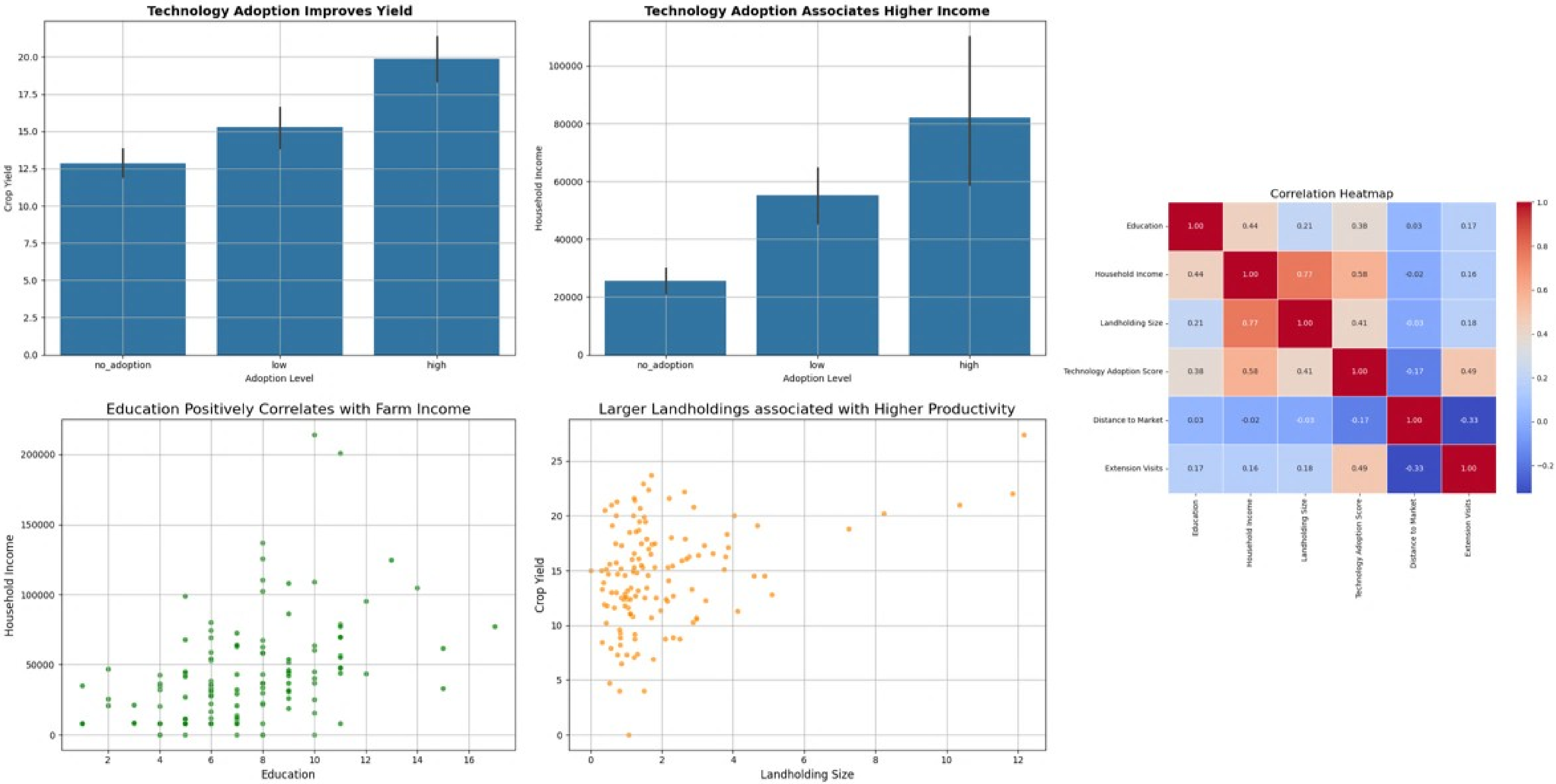
Data Analysis Tools and Software

The study uses Python for data processing, analysis, and visualization.
The following Python libraries were used:

- **Pandas** and **NumPy** – for data cleaning, manipulation, and numerical computations
- **re (Regular Expressions)** – for text processing and data standardization
- **RapidFuzz** – for handling inconsistent textual entries in raw data
- **Matplotlib** and **Seaborn** – for data visualization, including charts, graphs, and correlation plots
- **Statsmodels** – for estimating regression models and statistical inference
- **Scikit-learn (StandardScaler)** – for data scaling and preprocessing before analysis
- **PIL (Python Imaging Library)** – for handling and processing image-based outputs for report submission

Final AI Disclosure Statement

Generative AI tools were used in a limited capacity to assist with debugging minor Python code issues and automating repetitive tasks, such as the generation of batch scripts for folder structures. No AI-generated content was directly incorporated into the research analysis or results. All methodological decisions, interpretations, and conclusions are solely those of the author.


Technology Adoption Improves Yield
Crop Yield
Adoption Level
no_adoption
low
high
Technology Adoption Associates Higher Income
Household Income
Adoption Level
no_adoption
low
high
Correlation Heatmap
Education
Household Income
Landholding Size
Technology Adoption Score
Distance to Market
Extension Visits
1.00 0.44 0.21 0.38 0.03 0.17
0.44 1.00 0.77 0.58 -0.02 0.16
0.21 0.77 1.00 0.41 -0.03 0.18
0.38 0.58 0.41 1.00 -0.17 0.49
0.03 -0.02 -0.03 -0.17 1.00 -0.33
0.17 0.16 0.18 0.49 -0.33 1.00
Education Positively Correlates with Farm Income
Household Income
Education
Larger Landholdings associated with Higher Productivity
Crop Yield
Landholding Size

SUMMARY STATISTICS

	Education	Household Income	Landholding Size	Technology Adoption Score	Distance to Market	Extension Visits
count	8	8	8	8	8	8
mean	23.4	49737.85	19.8	18.26	23.95	17.58
std	46.15	69515.72	47.91	48.01	46.63	47.06
min	1	0.23	0	0	0.5	0
25%	5.21	9708.75	1.21	0	4.04	0
50%	7.18	35661.55	1.91	1.03	5.98	0.83
75%	11	45783.35	4.7	2.75	13.36	1.75
max	137	214100	138	137	138	134

LOGISTIC REGRESSION

	Coef.	Std.Err.	z	P>|z|	[0.025	0.975]
const	-2.65	1.65	-1.6	0.11	-5.9	0.59
age	-0.22	0.3	-0.72	0.47	-0.8	0.37
edu_years	1.15	0.44	2.59	0.01	0.28	2.01
land_acres	1.75	0.82	2.14	0.03	0.14	3.36
irrig_acres	0.01	0.44	0.03	0.97	-0.84	0.87
credit_acce	1.84	0.66	2.78	0.01	0.54	3.13
ext_visits	0.51	0.3	1.66	0.1	-0.09	1.1
training_da	0.31	0.34	0.92	0.36	-0.35	0.97
dist_mkt_k	-0.86	0.4	-2.15	0.03	-1.64	-0.08
gender_M	2.37	1.3	1.82	0.07	-0.18	4.92
caste_OBC	-0.5	1.44	-0.35	0.73	-3.32	2.32
caste_ST	-0.48	1.6	-0.3	0.76	-3.61	2.65
caste_SC	-1.26	1.6	-0.79	0.43	-4.38	1.87

LINEAR REGRESSION - YIELD

	Coef.	Std.Err.	z	P>|z|	[0.025	0.975]
const	14.55	0.46	31.97	0	13.66	15.44
tech_score	2.27	0.36	6.22	0	1.55	2.98
land_acres	0.04	0.64	0.06	0.95	-1.21	1.29
irrig_acres	0.65	0.41	1.59	0.11	-0.15	1.44
credit_acce	-1.91	0.8	-2.39	0.02	-3.47	-0.34
edu_years	0.23	0.36	0.65	0.52	-0.47	0.93
hh_size	-0.32	0.33	-0.98	0.33	-0.96	0.32
farm_exp_	0.83	0.31	2.69	0.01	0.23	1.44

LINEAR REGRESSION - INCOME

	Coef.	Std.Err.	z	P>|z|	[0.025	0.975]
const	9.59	0.37	25.93	0	8.86	10.31
tech_score	0.31	0.24	1.29	0.2	-0.16	0.77
land_acres	0.27	0.32	0.83	0.4	-0.36	0.9
irrig_acres	-0.1	0.27	-0.37	0.71	-0.64	0.43
credit_acce	0.81	0.43	1.9	0.06	-0.02	1.65
edu_years	0.33	0.18	1.82	0.07	-0.03	0.69
hh_size	0.09	0.21	0.43	0.67	-0.32	0.5
farm_exp_	-0.25	0.23	-1.06	0.29	-0.7	0.21